

Michael Ushman

Sr. Software Engineer | Sr. Python Engineer | Sr. Data Engineer

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Location: Buffalo, NY

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Work Authorization: US citizen

Professional Skills

- Experienced and results-driven **Python Full Stack Developer** with over 12 years of expertise in building scalable, efficient, and secure applications.
- Proven track record of developing complex web applications, **RESTful APIs**, and **ETL data pipelines** using frameworks like **Django**, **Flask**, **FastAPI**, and **FastAPI**.
- Extensive experience in building high-performance solutions using **Pandas**, **NumPy**, and modern workflow management systems like **Airflow** and **Flyte** for automating data pipelines at scale.
- Adept at creating **CLI tools**, and designing data processing workflows that enhance productivity and optimize backend processes.
- Strong problem-solving skills and an ability to adapt to evolving technologies in both frontend and backend development.

Technical Skills

- **Languages:** Python, JavaScript (React, Node.js), SQL, Bash
- **Web Technologies:** HTML5, CSS3, JavaScript, React, Django, Flask, FastAPI, RESTful APIs
- **Data Processing:** Pandas, NumPy, Matplotlib, Scikit-learn
- **Database Management:** MySQL, PostgreSQL, MongoDB, SQLite, Redis
- **ETL & Data Pipelines:** Airflow, Flyte, Apache Kafka, Apache Spark
- **Version Control:** Git, GitHub, GitLab
- **Cloud & DevOps:** AWS (Lambda, S3, EC2), Docker, Kubernetes, CI/CD (Jenkins, GitLab CI)
- **Data Visualization:** Plotly, Dash, Tableau
- **Testing:** PyTest, Unittest, Jest
- **Communication Protocols:** RESTful APIs, gRPC
- **Security:** Cognito for authentication and authorization
- **Tools:** Swig, Postman
- **Infrastructure:** Nginx, Linux, Ubuntu
- **Messaging Services:** Amazon SQS/SNS, NSQ
- **Database Integration:** Both relational and NoSQL

Professional Experience

Tata Consultancy Services, Philadelphia, PA

Apr 2022 - Present

Senior Python Full Stack Engineer

Responsibilities:

- Spearheaded the development of **REST APIs** using **FastAPI** for microservice-based applications, achieving a 40% improvement in performance over traditional Django-based approaches.
- Collaborated with the data engineering team to design robust **ETL pipelines** utilizing **Airflow** and **Flyte**, automating complex workflows that reduced processing time by 50%.
- Implemented **WebSocket** connections to enable real-time features like live chat and notifications.
- Integrated **Pandas** and **NumPy** to create efficient data transformation processes for real-time data analytics applications, ensuring that critical data was processed within tight SLAs.
- Designed and developed several backend systems using **Django** for building feature-rich applications, managing data access and ensuring security with authentication and authorization mechanisms such as **JWT** and **OAuth**.
- Led the migration of legacy systems to modern frameworks like **FastAPI** and **Django**, ensuring scalability and reducing maintenance complexity.
- Developed a range of **CLI tools** to automate data extraction, reporting, and validation, significantly improving team productivity and ensuring data consistency.
- Implemented **CI/CD pipelines** for efficient deployment using **Jenkins** and **GitLab CI**, streamlining the development lifecycle and improving delivery speed.
- Conducted thorough unit and integration testing using **PyTest** and **Jest**, ensuring the reliability and stability of the application.
- Optimized data storage and retrieval operations by leveraging **PostgreSQL** and **Redis**, ensuring high availability and fast query performance in production systems.
- Successfully led a team of developers and mentored junior engineers in best practices for Python development and data pipeline automation.
- Developed real-time data processing systems that integrate with third-party services using **AWS Lambda** and **S3** to handle streaming data and log analysis.
- Constantly improved system performance and scalability through profiling and optimization techniques, ensuring that the system could handle millions of users and data points.

Adobe, Columbia, SC
Senior Python Full Stack Developer

May 2019 – Jan 2022

Responsibilities:

- Designed and implemented **RESTful APIs** using **Django** to expose business logic to the frontend, improving integration with React-based web applications.
- Integrated **NumPy** and **Pandas** for data analysis and transformation, building data pipelines to process over a billion records for reporting purposes.
- Created and maintained **ETL pipelines** using **Airflow**, automating the extraction, transformation, and loading of data from diverse data sources to cloud-based storage systems like **AWS S3**.
- Collaborated with frontend teams to implement features like real-time data updates using **Web-Sockets** and optimized data-fetching strategies using **GraphQL**.
- Integrated data storage solutions including **PostgreSQL** and **MongoDB**, ensuring data integrity, security, and fast data retrieval.
- Automated system deployment and scaling using **Docker** and **Kubernetes**, reducing downtime during system upgrades and handling high traffic volumes.
- Developed automated scripts to handle backup, monitoring, and alerting for critical database systems.
- Utilized **AWS Sagemaker** for model deployment, creating scalable and efficient **AI** services for customer-facing applications.
- Built **command-line interfaces (CLI)** for internal use to streamline data processing and reporting tasks, saving the team hours of manual work each week.
- Worked closely with clients to translate business requirements into technical solutions, ensuring that the software met both functional and non-functional requirements.
- Trained junior developers on Python best practices, improving code quality and team efficiency.
- Introduced performance monitoring and logging solutions using **AWS CloudWatch** and **Prometheus** to proactively address performance bottlenecks.
- Worked on **data migration** projects, moving client data from on-premise systems to cloud-based architectures, ensuring minimal downtime and secure transfer.

Mphasis, San Ramon, CA
Python Engineer

Oct 2013 – Mar 2019

- Played a key role in version control, continuous integration, and configuration management using GIT, Jenkins, Docker, and Chef.
- Developed and deployed **Flask-based REST APIs** that powered several web applications, handling hundreds of requests per second and ensuring high availability.

- Built robust backend systems for managing customer data, transactions, and logs in **MySQL** and **PostgreSQL**, ensuring security and scalability.
- Worked with frontend developers to implement authentication systems using **JWT** for stateless sessions and secure access to resources.
- Implemented database indexing and query optimization strategies to handle large amounts of transactional data and improve response times.
- Introduced automated **unit and integration testing** using **PyTest** to ensure the reliability of backend services and improve code quality.
- Optimized data workflows using **Pandas** to perform batch processing and aggregation of customer data, reducing processing time by 35%.
- Collaborated with the DevOps team to automate deployment processes and implement **CI/CD pipelines** using **Jenkins** and **Docker**.
- Integrated external services using **REST APIs** and **web scraping** techniques to enrich data and provide additional functionalities in the application.
- Participated in code reviews and mentoring junior developers on Python and software engineering best practices.
- Developed **CLI tools** to automate daily data import tasks and system monitoring, improving team productivity.
- Implemented server-side caching strategies using **Redis** to reduce load times and improve user experience.
- Wrote technical documentation and API specs to ensure clarity and consistency across all services and endpoints.

Education

University of South Carolina Major in Computer Science (2008 – 2012)

- Software engineering
- Artificial Intelligence & Mathematics and statistics
- Algorithm
- Database System
- GPA: 3.8